



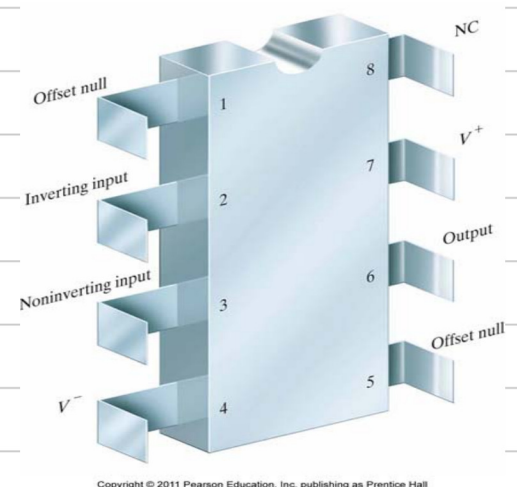
Chapter 13, Operational Amplifier

An operational amplifier (op amp) is an electronic circuit that contains many electronic devices such as transistors and diodes. The detailed analysis of op amp requires an understanding of electronic devices which is beyond the scope of this course. However, it can be used as a circuit building block by focusing on its terminal behavior.

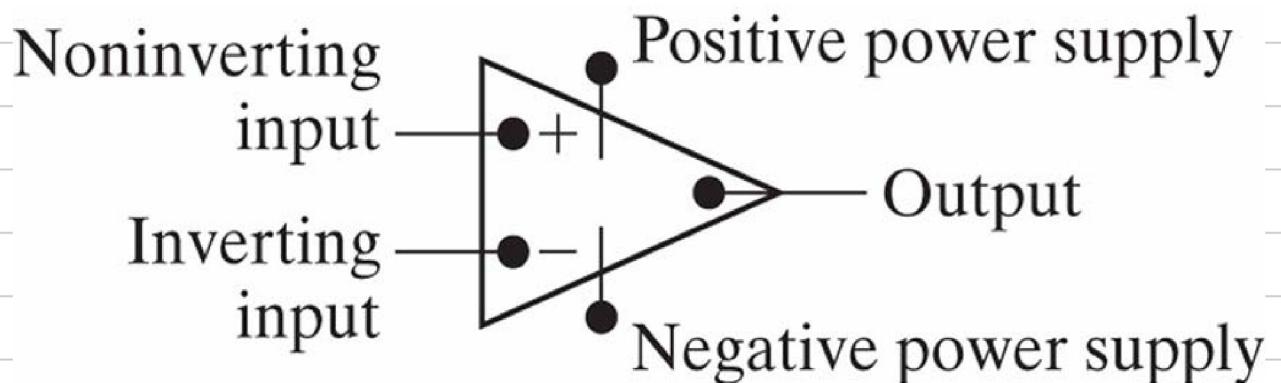
The operational amplifier came into existence as a basic building block for analog computers. It was referred to as operational because it was used to implement the mathematical operations such as addition, sign changing, comparing, integration and differentiation.



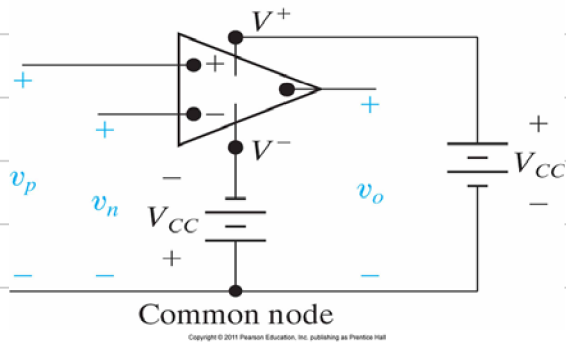
A typical Op amp Package



pin configuration



circuit symbol



range of the output voltage:
 $V^- < V_o < V^+$

terminal voltage connection

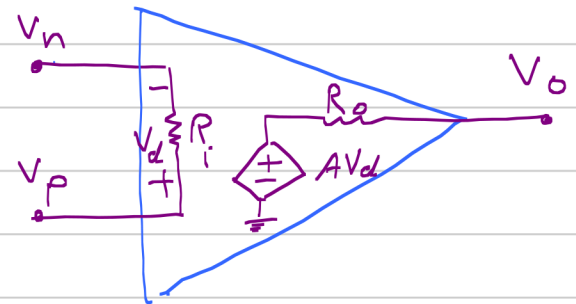
The equivalent circuit for non-ideal op amp

R_i : input resistance (large: $10^6 - 10^{13} \Omega$)

R_o : output resistance (small: $10 - 100 \Omega$)

A : open loop gain (large: $10^5 - 10^8$)

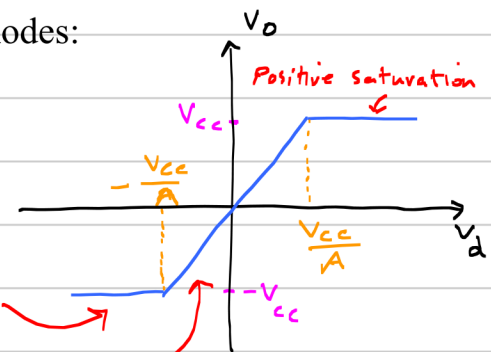
V_d : differential input, $V_d = V_p - V_n$



$$V_o = A(V_p - V_n)$$

voltage transfer characteristics

without a feedback connection, op amp can operate in three modes:



$$V_o = \begin{cases} -V_{cc} & \text{if } AV_d < -V_{cc}, \text{ (negative saturation)} \\ AV_d & \text{if } -V_{cc} < AV_d < V_{cc}, \text{ (Linear region)} \\ V_{cc} & \text{if } AV_d > V_{cc}, \text{ (positive saturation)} \end{cases}$$

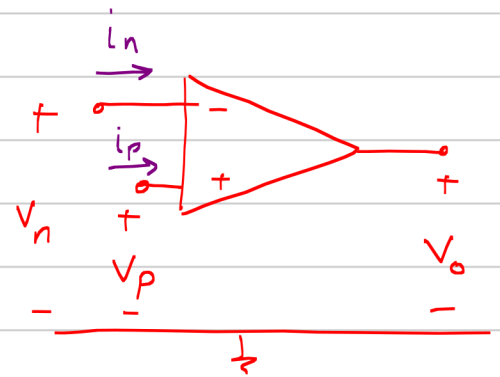
In most practical applications, the output voltage of the op amp is fed back to the inverting input. This is called a negative feedback.

When a negative feed back is provided, the ratio of the output to the input voltage is called the closed loop gain which is insensitive to the open loop gain A .

Ideal op amp

characteristics of an ideal op amp :

1. infinite open loop gain , $A = \infty$
2. infinite input resistance , $R_i = \infty$
3. zero output resistance , $R_o = 0$



Due to the infinite input resistance, an open circuit exists at the input which implies zero input current.

$$i_p = 0 , \quad i_n = 0$$

Due to infinite open loop gain , the differential voltage $V_d = V_p - V_n$ is zero.

$$V_o = A(V_p - V_n) \Rightarrow V_p - V_n = \frac{V_o}{A} = 0 \Rightarrow V_p = V_n$$

∞

Inverting amplifier

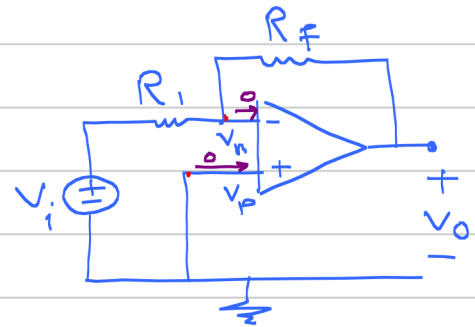
$$V_p = 0 \Rightarrow V_n = 0$$

$$\frac{0 - V_i}{R_i} + \frac{0 - V_o}{R_f} + 0 = 0$$

$$V_o = -\frac{R_f}{R_i} V_i$$

$$\boxed{\frac{V_o}{V_i} = -\frac{R_f}{R_i}}$$

An inverting amplifier reverses the polarity of the input signal and amplifies it by the gain of $\frac{R_f}{R_i}$.



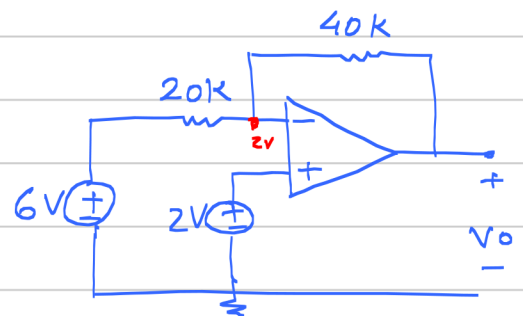
Ex. a) Find V_o .

b) If $V_{cc} = \pm 5V$, what is V_o ?

a) $V_p = 2V \rightarrow V_n = 2V$

$$\frac{2 - 6}{20k} + \frac{2 - V_o}{40k} + 0 = 0$$

$$\boxed{V_o = -6V}$$



b) If the power supply is connected to $\pm 5V$, the op amp will operate in the negative saturation region because V_o can not exceed the power supply voltages.

$$-5 \leq V_o \leq 5 \Rightarrow \boxed{V_o = -5V}$$

Non-inverting amplifier

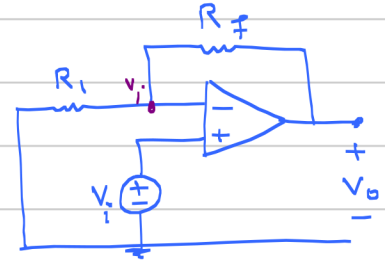
$$V_p = V_i \rightarrow V_n = V_i$$

$$\frac{V_i}{R_i} + \frac{V_i - V_o}{R_f} + 0 = 0$$

$$V_i \left(\frac{1}{R_i} + \frac{1}{R_f} \right) = \frac{V_o}{R_f} \Rightarrow$$

$$V_o = \left(1 + \frac{R_f}{R_i} \right) V_i$$

The output voltage polarity is the same as the input voltage and the gain is $1 + \frac{R_f}{R_i}$.

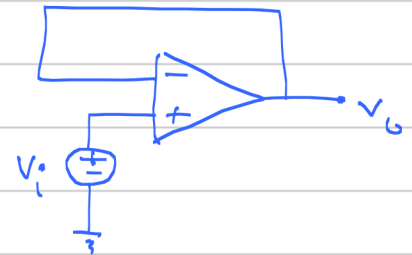


Unity gain amplifier (voltage follower)

$$V_o = V_n = V_p = V_i$$

$$\Rightarrow \boxed{V_o = V_i}$$

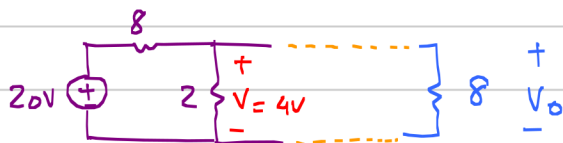
so, what is it used for?



The voltage follower is used as a buffer, to isolate the input ckt. from the output circuit and eliminate the loading effect.

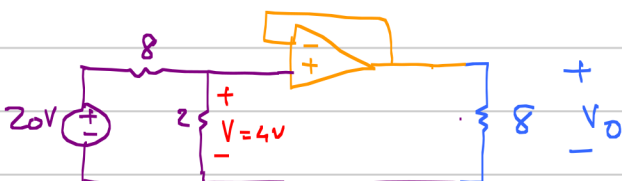
Example

Direct connection:

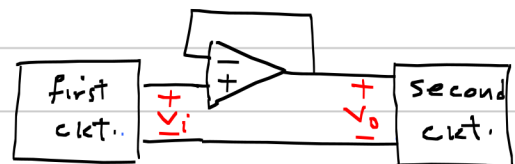


$V_o \neq 4V$ (because of the loading effect)

Connect via a buffer:



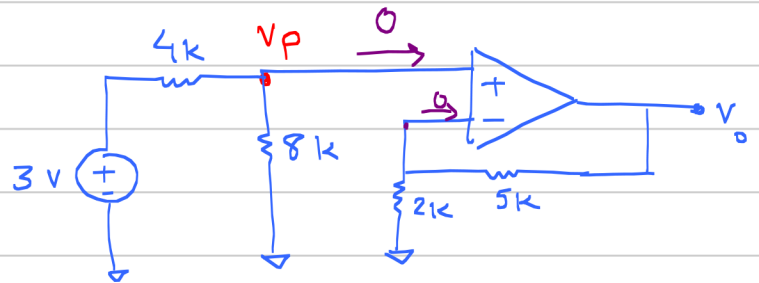
$$V_o = V = 4V$$



$$V_o = V_i$$

Ex. Find V_o .

Since $V_p = 0$, $4k$ and $8k$ are in series. VDR can be applied.



$$V_p = 3 \frac{8}{8+4} = 2V$$

$$\Rightarrow V_n = 2V$$

$$\frac{2}{2k} + \frac{2-V_o}{5k} + 0 = 0 \Rightarrow \boxed{V_o = 7V}$$

or, using VDR:

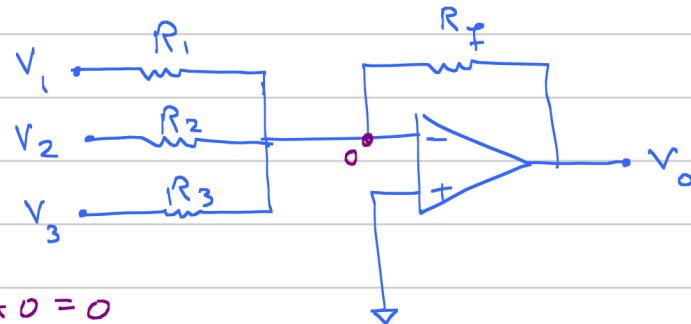
$$V_n = V_o \frac{2}{2+5}$$

$$2 = V_o \frac{2}{7}$$

$$\underline{V_o = 7}$$

Summing amplifier

$$V_p = 0 \rightarrow V_n = 0$$



$$\frac{0-V_1}{R_1} + \frac{0-V_2}{R_2} + \frac{0-V_3}{R_3} + \frac{0-V_o}{R_f} + 0 = 0$$

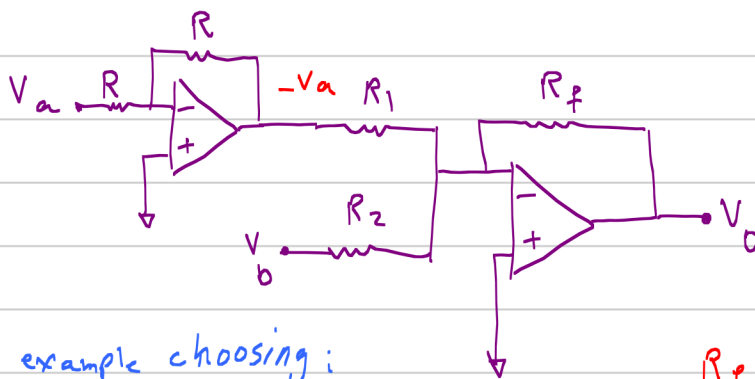
$$-\left(\frac{1}{R_1} V_1 + \frac{1}{R_2} V_2 + \frac{1}{R_3} V_3\right) = \frac{V_o}{R_f}$$

$$\boxed{V_o = -\left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3\right)}$$

This ckt. Combines the input signals and produces an output which is the sum of the weighted input signal with the inverted sign.

Ex. Design an op amp circuit to perform the following operation: $V_o = 3V_a - 2V_b$

First we need to invert V_a . Then input $-V_a$ and V_b into a summing amplifier.



$$V_o = -\left(\frac{R_f}{R_1}(-V_a) + \frac{R_f}{R_2}V_b\right)$$
$$= \underbrace{\left(\frac{R_f}{R_1}\right)}_3 V_a - \underbrace{\left(\frac{R_f}{R_2}\right)}_2 V_b$$

for example choosing:

$$R_1 = 20\text{ k}$$

$$R_2 = 30\text{ k}$$

$$R_f = 60\text{ k}$$

$$R_f = 3R_1 = 2R_2$$